

Exp No: 13 EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

Github : <https://github.com/Pardhav2006/embedded-system>

Program:

```
#include <reg51.h>

void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
    unsigned long sum = 0;
    for(i = 0; i < 5; i++)
    {
        sum = sum + array[i];
    }
    P0 = (unsigned char)(sum & 0xFF);    // Lower 8 bits
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Middle 8 bits
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
    while(1); // Keep output stable
```

}

C:\Users\yadah\OneDrive\Documents\embedded\EXP 13.uxproj - µVision

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Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp_max	0x15
dptr	0x0000
PC	0x0909
status	309
sec	0.000422s
psw	0x00

Disassembly

```
3: void main(void)
4: {
5:     unsigned int i;
6:     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7: }
```

exp13.c

```
1#include <reg51.h>
2
3void main(void)
4{
5    unsigned int i;
6    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7    unsigned long sum = 0;
8
9    for(i = 0; i < 5; i++)
10    {
11        sum = sum + array[i];
12    }
13
14    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
15    P1 = (unsigned char)((sum >> 8) & 0xFF); // Middle 8 bits
16    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
17
18    while(1); // Keep output stable
19}
```

Parallel Port 0

Port 0	7	6	5	4	3	2	1	0
P0	0xCC	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Pins	0xCC	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Parallel Port 1

Port 1	7	6	5	4	3	2	1	0
P1	0xAB	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Pins	0xAB	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Parallel Port 2

Port 2	7	6	5	4	3	2	1	0
P2	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Pins	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Watch 1

Name	Value	Type
sum	0x0001ABCC	ulong
array[0]	0x1111	uint
array[1]	0x2222	uint
<Enter expression>		

Command Call Stack + Locals Watch 1 Memory 1

Simulation T1: 40.99319661 sec L4 C.2 CAP. NUM. SCL. OVR. B/W